

Keyu Wang

☎ (+49) 1525 6272814 | ✉ keyu.wang@student.uni-tuebingen.de | 🌐 website | 🎓 Google Scholar | 🐙 GitHub | 🔗 LinkedIn

EDUCATION

University of Tuebingen

M.Sc. in Machine Learning

Tuebingen, Germany

Oct. 2024 – Present

Current GPA: 1.6 (German scale; 1.0 = best, 5.0 = worst)

Relevant Coursework: Non-convex Optimization for Deep Learning, Statistical Machine Learning, Probabilistic Machine Learning, The Science of Machine Learning Benchmarks, etc.

Southeast University

B.Eng. in Artificial Intelligence

Nanjing, China

Sep. 2020 – Jun. 2024

Cumulative GPA: 3.79/4.0

Relevant Coursework: Optimization, Probability Statistics & Stochastic Processes, Distributed Systems, Machine Learning, Natural Language Processing, Reinforcement Learning, etc.

PUBLICATIONS

- [1] **Keyu Wang***, Yangyi Huang*, Jiale Kang, David González-Martínez, Weiyang Liu, and Shiwei Liu. “DepthBench: Measuring How Residual Connections Enable More Computational Depth.” **Under Review, arXiv 2026**.
- [2] Di He, Songjun Tu, **Keyu Wang**, Lu Yin, and Shiwei Liu. “One LR Doesn’t Fit All: Heavy-Tail Guided Layerwise Learning Rates for LLMs.” **ICML 2026**.
- [3] **Keyu Wang***, Tian Lyu*, Guinan Su, Lu Yin, Marco Canini, Jonas Geiping, and Shiwei Liu. “When Fewer Layers Break More Chains: Layer Pruning Harms Test-time Scaling in LLMs.” **COLM 2026**.
- [4] Xinyuan Song, **Keyu Wang**, Pengxiang Li, Lu Yin, and Shiwei Liu. “Demystifying the Roles of LLM Layers in Retrieval, Knowledge, and Reasoning.” **Technical Report, arXiv 2025**.
- [5] **Keyu Wang**, Guilin Qi, Jiaqi Li, and Songlin Zhai. “Can Large Language Models Understand DL-Lite Ontologies? An Empirical Study.” **Findings of EMNLP 2024**.
- [6] **Keyu Wang**, Site Li, Jiaye Li, Guilin Qi, and Qiu Ji. “An Embedding-based Approach to Inconsistency-tolerant Reasoning with Inconsistent Ontologies.” **IJCKG 2023**.

RESEARCH EXPERIENCE

Max Planck Institute for Intelligent Systems & ELLIS Institute Tuebingen

Research Assistant, WEI Lab | Advisor: Dr. Shiwei Liu

Tuebingen, Germany

Mar. 2025 – Present

Measuring Residual Connections for Computational Depth:

Lead; 📄 **Preprint 2026**

- Developed DEPTHBENCH to measure how residual connection design affects computational depth, by systematically varying the width–depth aspect ratio ($d_{\text{model}}/n_{\text{layer}}$) from shallow–wide to deep–narrow shapes, while keeping the model size and pre-training recipe fixed across 10 representative architectures.
- Revealed that allocating more capacity to depth is strongly architecture-dependent: standard Pre-LN and most norm- and scaling-based variants yield little benefit and can even degrade performance as models grow deeper–narrower, whereas HC and Full AttnRes improve consistently at extreme depth.
- Established a suite of controlled layer-level analyses, showing that the gains of HC and Full AttnRes are associated with more effective layer utilization and distinct mechanisms of computational depth.

Pre-training with Layerwise Learning Rates:

Co-author; 📄 **ICML 2026**

- Developed an adaptive pretraining scheme with layerwise learning rates guided by Heavy-Tail Theory to promote balanced training across layers, faster convergence with improved generalization.

Evaluating Layer Pruning for Test-time Scaling:

Lead; 📄 **COLM 2026**

- Investigated how layer pruning affects reasoning under sequential and parallel test-time scaling, and whether supervised fine-tuning can recover the lost capabilities.

- Challenged the widely circulated yet potential misleading claim “layer pruning is a free lunch”, highlighting that even mild pruning degrades test-time scaling by inducing recurring loops, reducing solution diversity, and weakening self-correction; standard SFT only partially recovers these losses.

Demystifying the Roles of LLM Layers:

Co-author;  **Preprint 2025**

- Conducted a systematic analysis of layer roles in retrieval, knowledge, and reasoning across evaluation protocols, task categories, and model families.
- Revealed that LLM depth usage is highly heterogeneous: deeper layers are dispensable under likelihood-based metrics but indispensable for reasoning and long-range coherence under generation.

Max Planck Institute for Intelligent Systems & ELLIS Institute Tuebingen Tuebingen, Germany
Master’s Student, SEAL Lab | Advisor: Dr. Jonas Geiping Nov. 2025 – May 2026

Developing Hierarchical KV Cache Compression for Efficient Reasoning

Lead

- Developed a hierarchical KV-cache compression method that recursively summarizes token chunks into multi-level representations under a fixed cache budget while preserving performance on commonsense and knowledge-reasoning tasks.

Southeast University

Undergraduate Student, COIN Lab | Advisor: Prof. Dr. Guilin Qi

Nanjing, China
Feb. 2022 – Jun. 2024

Evaluating LLMs with Description Logics:

Lead;  **EMNLP 2024 Findings**

- Evaluated LLMs’ understanding of Description Logics across six tasks: syntax checking, instance checking, subsumption checking, functionality checking, query answering, and satisfiability checking.
- Found that LLMs capture formal syntax and the model-theoretic semantics of concepts and roles, but struggle with negative inclusions and deduction over large instance sets.

Inconsistency-tolerant Reasoning in Description Logics:

Lead;  **IJCKG 2023**

- Proposed a monotonic selection relation for ranking maximal consistent subsets of inconsistent belief bases, extending from propositional logic to description logics using semantic embeddings.
- Proved key logical properties required for rational inference and empirically demonstrated an improvement of more than 10% in reasoning performance.

SERVICE

Conference Reviewer: COLM 2026, CPAL 2026

Workshop Reviewer: On-device Intelligence @ NeurIPS 2026

SKILLS

Programming: Python (PyTorch), C/C++.

Expertise: LLM Pretraining, SFT, Inference & Evaluation

Languages: Chinese (Native), English (Fluent, IELTS 7.5/C1).